

Shreyaa Sanjay

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EDUCATION

Cornell University

Ithaca, NY

Bachelor of Science in Computer Science, Minor in Electrical and Computer Engineering

Expected May 2028

Relevant Coursework: Intro to Operations Research, Development in Python, Full Stack Web Development, Object Oriented Programming and Data Structures, Discrete Math, Linear Algebra, Functional Programming in OCaml, Digital Logic and Embedded Systems in Verilog

Campus Involvement: Resident Advisor, Cornell Hack4Impact, Women in Computing at Cornell

WORK EXPERIENCE

Cornell Electric Vehicles

Ithaca, NY

Software Engineer

September 2025 – Present

- Built a JavaScript/React + FastAPI dashboard to visualize 10+ live sensor streams for real-time vehicle diagnostics.
- Architected a ROS2 telemetry system over Starlink + Tailscale VPN, configuring FastDDS data transport to achieve end-to-end sensor latency under 30ms from shared-memory buffers to live subscriber databases; used by 70+ team members
- Developed live dashboard visualizations for YOLO bounding boxes, SLAM algorithms, and simulated vehicle output; system earned 1st Place in Data and Telemetry at the Shell Eco-marathon.

CS3 (Center for Smart StreetScapes, WINLAB)

New Brunswick, NJ

Machine Learning Engineering Intern

May 2026 – August 2026

- Designed a multi-agent system that answers live questions by generating lightweight workflows for smart-city applications.
- Built a 6-stage agent workflow and user interface for live inference from a 4-camera smart space, connecting retrieval and answer agents through the runtime, achieving a 87.8% accuracy after 50 trials.
- Formally verified 120+ agent workflows using PlusCal/TLA+ and TLC, repairing 23 invalid protocols before execution.
- Built a Docker-based LLM harness with tool-calling and API integrations to extract web data under context-window constraints while reducing tool calling by 2x using a cache classifier for intent decomposition of the prompt.

Bonassar Engineering Lab

Ithaca, NY

Computer Science Research Assistant

September 2025 – Present

- Developed an interactive GUI with 2 real-time graphs and CSV logging for impedance, PID, and motor-control data.
- Designed and assembled a custom PCB integrating a stepper-motor driver with WaveForms-based impedance sensing
- Unified Raspberry Pi Pico connectivity and PID-control output in one program, closing the loop from impedance sensing through a stepper-motor driver to extrusion-rate adjustment.
- Tested time alignment between impedance readings and motor speed, supporting reproducible experiments across bioinks.
- Analyzed Likert-scale survey data from 3 senior BME courses; creating violin plots for 18 sub-questions to evaluate learning.

PROJECTS

EgoMemory for Ego-Exo4D | *Python, CLIP, FAISS, Tkinter*

[GitHub](#)

- Built a multimodal episodic-memory retrieval system for egocentric video using CLIP visual embeddings, aligned transcripts, temporal context, and trajectory/motion features with FAISS retrieval.
- Conducted a modality experiment across 50 manually validated queries from Ego-Exo4D; full multimodal retrieval achieved 84% Recall@5 versus 30% for vision-only and 18% for motion-only retrieval.
- Developed a native desktop GUI displaying ranked memory windows, modality evidence, synchronized captions, and retrieved video segments for qualitative analysis.

CerviCheck Diagnosis Device | *Python, Tkinter, PySerial, Matplotlib, C++, STM32*

[GitHub](#)

- Built a Python/Tkinter and PySerial control GUI for pressure-sweep configuration, calibration, and bidirectional serial data collection from a cervical-tissue testing device.
- Developed real-time visualization and Cauchy stress-strain fitting for pressure, deformation, and pad-contact signals; supported bench tests distinguishing 1–3% differences in gel mechanical properties.
- Integrated hydration sensing with a 16-pad gold-copper impedance array and ported Arduino C++ pressure-control logic to an STM32 microcontroller; collaborated with OB-GYN physicians on clinician-facing usability.

TECHNICAL SKILLS

Languages: Java, Python, C++, Verilog, SQL, JavaScript, HTML/CSS, OCaml

Frameworks/Libraries: React, Vite, FastAPI, NumPy, Matplotlib, LangGraph, ROS2, TLA+

Developer Tools: Git, Linux, Docker, PostgreSQL